

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

## 2 DATA AND METHODOLOGY

We draw data from the O\*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O\*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O\*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-

veys. To ensure compatibility with standard occupational classifications, we translated these estimations to the SOC-18 classification using O\*NET’s crosswalks for a total of 759 occupations.

To enhance our understanding of the GENOE’s effectiveness and to validate our methodology, we conducted several additional exercises. Firstly, we measured task-level exposure by assessing the proportion of tasks at risk within each occupation, allowing us to identify roles where AI might significantly impact certain tasks without necessarily replacing the entire job. We also evaluated tasks in isolation and compared the results with our holistic occupation-level approach to understand the value of considering tasks within their broader occupational context. Additionally, we tested the effect of withholding the occupation name from the model to gauge the importance of occupational context in the AI’s assessment. To incorporate broader societal considerations, we explicitly prompted the model to consider ethical and social factors in its assessments.

Furthermore, we conducted follow-up surveys, asking the AI to explain its reasoning behind the exposure estimates, effectively “opening the black box” of its decision-making process. These additional analyses serve multiple purposes: they validate our methodology by providing alternative perspectives on AI exposure, offer insights into the factors influencing AI’s potential impact on various occupations, and help us understand the decision-making process of the AI model. This approach not only strengthens the robustness of our GENOE index but also potentially reveals any biases or limitations in the AI’s assessments, providing a more reliable picture of occupational exposure to AI.

To further validate our approach, we applied our synthetic survey methodology to replicate the Felten *et al.* (2021) index of occupational exposure. In this exercise, we substituted the original human expert surveys with synthetic surveys conducted using ChatGPT4o. The primary objective was to assess the capability of our AI-driven methodology to reproduce results comparable to those obtained through traditional human expert assessments. This replication serves as a crucial test of our approach’s reliability and accuracy, offering insights into the potential of large language models to emulate human expertise in evaluating occupational exposure to AI.

### 3 RESULTS

#### 3.1 Baseline Estimations

Figure 1 presents a comprehensive view of AI exposure across 759 occupations, illustrating the average, median, 25th percentile, and 75th percentile for the three defined time horizons. The analysis reveals a progressive increase in average AI exposure: 0.28 within one year, rising to 0.38 over five years, and reaching 0.44 over a decade. This pattern suggests that as AI technologies advance and mature, their potential to reshape various professions grows significantly.<sup>5</sup>

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<sup>5</sup>Complete descriptive statistics can be found in Appendix Table A.1.

**Table 4: Descriptive Statistics of Alternative Methodologies for GENOE**

|                               | Baseline GENOE | Blind to occup. name | Task-by-task | Ethical emphasis | Ethical exclusion | LLMs   | Non-AI technologies | Only AI |
|-------------------------------|----------------|----------------------|--------------|------------------|-------------------|--------|---------------------|---------|
| mean                          | 0.382          | 0.391                | 0.560        | 0.370            | 0.423             | 0.204  | 0.343               | 0.039   |
| p25                           | 0.245          | 0.283                | 0.446        | 0.238            | 0.277             | 0.100  | 0.216               | 0.005   |
| p50                           | 0.317          | 0.325                | 0.546        | 0.312            | 0.355             | 0.175  | 0.295               | 0.031   |
| p75                           | 0.475          | 0.469                | 0.680        | 0.438            | 0.594             | 0.261  | 0.399               | 0.068   |
| min                           | 0.095          | 0.117                | 0.250        | 0.098            | 0.079             | 0.009  | 0.089               | -0.108  |
| max                           | 0.932          | 0.862                | 0.891        | 0.885            | 0.945             | 0.857  | 0.919               | 0.391   |
| Corr with baseline            | 1.000          | 0.947                | 0.888        | 0.992            | 0.982             | 0.475  | 0.946               | 0.346   |
| Corr with FN                  | -0.081         | -0.043               | 0.065        | -0.102           | -0.082            | 0.661  | -0.261              | 0.507   |
| Corr with PZ                  | 0.569          | 0.559                | 0.616        | 0.560            | 0.564             | 0.666  | 0.430               | 0.511   |
| Corr with WB                  | 0.072          | 0.061                | 0.066        | 0.061            | 0.062             | -0.036 | 0.048               | 0.085   |
| Corr with FO                  | 0.545          | 0.501                | 0.470        | 0.546            | 0.553             | 0.029  | 0.613               | -0.082  |
| P-value of diff with baseline | .              | 0.000                | 0.000        | 0.000            | 0.000             | 0.000  | 0.000               | 0.000   |

*Source:* Authors’ calculations using data from O\*NET and surveys conducted with GPT-4o through OpenAI’s API. *Note:* This table shows the descriptive statistics of the alternative indexes at the SOC-2010 occupational level for the five-year horizon, the correlation with the baseline GENOE and other existing indexes, and the p-value of the mean difference with the baseline GENOE. FN refers to Felten *et al.* (2021), PZ to Pizzinelli *et al.* (2023), WB to Webb (2019) and FO to Frey and Osborne (2017).

These alternative methodologies serve a dual purpose: they validate the robustness of our baseline GENOE index while offering deeper insights into the complex landscape of AI’s potential impact on the workforce. By examining occupational exposure through various lenses—from task-level granularity to ethical considerations and AI-specific advancements—we gain a more comprehensive understanding of the challenges and opportunities that lie ahead. Our findings underscore the necessity of a multifaceted approach in assessing the future of work, one that considers not only the technical capabilities of AI but also the contextual, ethical, and broader technological factors that shape occupational vulnerability. This perspective is crucial for policy-makers, employers, and workers as they navigate the evolving dynamics of human-AI interaction in the workplace and prepare for a future where the boundaries between human and artificial intelligence continue to shift.

### 3.4 Replication of Felten *et al.* (2021)’s AIOE Using Synthetic Surveys

To validate the effectiveness of synthetic surveys in producing assessments comparable to those of human experts, we replicated the well-established AI Occupational Exposure Index (AIOE) developed by Felten *et al.* (2021), which has been widely used in various empirical applications

(Acemoglu *et al.*, 2022; Fossen and Sorgner, 2021).<sup>7</sup> In our replication, we replaced the original human expert surveys with synthetic surveys conducted using GPT-4o via OpenAI’s API.

The AIOE measures the degree of overlap between advancements in AI and workplace abilities. They define 10 key AI applications from the Electronic Frontier Foundation, which have seen considerable development since 2010: abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, and speech recognition. They then link these AI applications to the 52 labor abilities recorded in O\*NET by conducting 2,000 surveys with gig workers from Amazon’s Mechanical Turk (MTurk).<sup>8</sup> Each participant was asked whether they believe each application is related to each ability, with a yes or no answer. These responses were processed as binary variables and averaged across all respondents to construct a matrix of relatedness, with a score between 0 and 1 for each application and each ability. The ability-level exposure was calculated as the sum of the scores of all applications for Ability  $j$  ( $A_j$ ). Finally, they calculated the occupation-level ( $AIOE_k$ ) for every occupation  $k$  as follows:

$$AIOE_k = \frac{\sum_{j=1}^{52} A_j \times L_{jk} \times I_{jk}}{\sum_{j=1}^{52} L_{jk} \times I_{jk}} \quad (1)$$

Using the replication files for the AIOE made available online by the authors, we re-estimated the  $AIOE_k$  by replacing the matrix built with mTurk surveys with a vector resulting from surveys conducted with GPT-4o through OpenAI’s API. Instead of defining the 10 AI applications, we asked the language model the following: “Based on recent advancements in AI, please answer ‘Yes’ or ‘No’ depending on whether you believe that AI is related to or could be used for the following ability: [ability name] [ability description].” This was done for the 52 abilities defined in O\*NET and repeated 100 times, with the temperature setting randomized. Although this approach differs from the original methodology by not specifying 10 applications, we made this change to minimize conditioning the model. This method likely results in a more conservative estimate of the index by reducing potential similarities between the original and replicated indexes.

Since we do not have specific AI applications, we calculated the  $A_j$  as 10 times the average score from the 100 iterations. This adjustment was made solely for comparability purposes and does not affect the index calculation. We then ran the same equation (1) and computed the new index.

The results from the estimation, and the comparison with the original index, can be observed in Figure 3. The correlation between both indexes is 0.99, indicating a very high degree of replicability and very small variations between the answers from the mTurk workers and the answers

<sup>7</sup>The AIOE was initially introduced in Felten *et al.* (2018) and subsequently modified and updated in Felten *et al.* (2021).

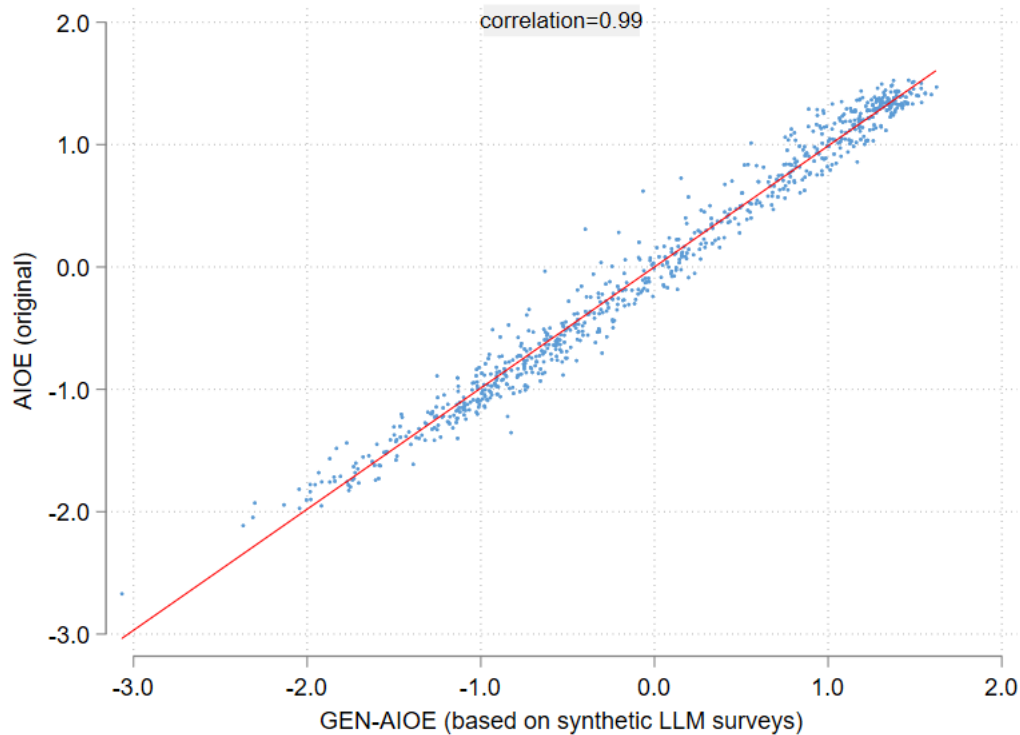
<sup>8</sup>Amazon Mechanical Turk (MTurk) is an online crowdsourcing platform that allows businesses to outsource a wide range of tasks to a global, distributed workforce. These tasks can vary from data validation and research to content moderation and survey participation.

provided by the synthetic surveys with GPT-4o.

A potential concern might be that GPT-4o could generate similar responses to mTurk workers because it searches the web or because its training data includes information from the AIOE and the mTurk surveys. However, GPT-4o does not perform real-time web searches unless explicitly instructed to do so, and in our study, we did not prompt GPT-4o to search the web. Regarding the second concern of the training data, as shown in follow-up surveys from the GENOE, the reasoning of the model is more holistic, based on its understanding of AI advancements and specific labor and workplace tasks (in this case, abilities), and does not rely on specific unique sources. Therefore, while it is possible that the training data could include some pre-existing information on Felten *et al.* (2021)'s work, among millions of other sources, the responses are synthesized from a broader knowledge base and not directly copied from any particular source.

Therefore, the high correlation observed suggests that the model's responses are consistent with human judgment for this specific task. This consistency supports the robustness and reliability of using artificial intelligence tools to emulate human analytical capabilities in these types of assessments.

**Figure 3: Correlation of Original AIOE and AIOE Built with Synthetic Surveys**



*Source:* Authors' formulation based on data from O\*NET and Felten *et al.* (2021) *Note:* Each dot represents an occupation at the SOC 10-digit code.